

SORO Bedionita

Curriculum Vitae

sorobedio@kaist.ac.kr (+82) 010-8857-6416

[Google Scholar](#) [LinkedIn](#)

Kim Jaechul Graduate School of Artificial Intelligence, KAIST

(Yangjae Industry-Academic Campus) 108, Taebong-ro, Seocho-gu, Seoul, Republic of Korea

Research Summary

I am currently a Postdoctoral Researcher at KAIST AI. My research focuses on generative weight-space learning, framing neural network optimization as a manifold learning problem. Through latent-space optimization, I leverage compressed knowledge from pretrained models to synthesize high-performance initializations and enable efficient manifold fine-tuning. My broader goal is to build a generative AutoML system that substantially reduces training costs by reusing knowledge embedded in pretrained weights to generate task-ready models for new architectures with little or no additional training.

Appointments

Postdoctoral Researcher

Feb. 2026 – Present

KAIST AI, Kim Jaechul Graduate School of AI

Seoul, South Korea

- Generative weight-space learning and weights latent-space manifold learning foundation models generation and optimization, merging, and efficient generative initialization.

Education

Ph.D. in Artificial Intelligence

Sept. 2020 – Feb. 2026

KAIST AI

Seoul, South Korea

- Supervisors: Song Chong and Sung Ju Hwang.
- Thesis: Weight-Space Learning as a Foundation for Deep AutoML: Efficient Knowledge Transfer and Model Generation.

M.Sc. in Electrical and Computer Engineering

Sept. 2017 – Aug. 2019

Ajou University

Suwon, South Korea

- Supervisor: Chae Woo Lee.
- Thesis: Indoor positioning and navigation based on deep neural networks and Wi-Fi fingerprinting.

Engineering Degree in Electronic Engineering

Sept. 2011 – Sept. 2014

Institut Polytechnique Felix Houphouet Boigny

Côte d'Ivoire

- Polytechnic engineering degree in Electronics and Computer Engineering.

Preparatory Classes (MPSI–MP)

Sept. 2009 – Sept. 2011

Institut Polytechnique Felix Houphouet Boigny

Côte d'Ivoire

- Intensive coursework in mathematics and physics.

Research Publications

1. **B. Soro** *et al.*, LS-Merge: Merging Language Models in Latent Space, submitted to *International Conference on Learning Representations (ICLR)*, 2026.
2. **B. Soro**, B. Andreis, S. Chong, and S. J. Hwang, Instruction-Guided Autoregressive Neural Network Parameter Generation, in *ICLR 2025 Workshop on Weight Space Learning (WSL)*.

3. **B. Soro**, B. Andreis, H. Lee, S. Chong, F. Hutter, and S. J. Hwang, Diffusion-based Neural Network Weights Generation, in *International Conference on Learning Representations (ICLR)*, 2025.
4. B. Andreis, **S. Bedionita**, and S. J. Hwang, Set-based Neural Network Encoding Without Weight Tying, in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
5. **B. Soro** and C. Song, Enhancing Differentiable Architecture Search: A Study on Small Number of Cell Blocks in the Search Stage, and Important Branches-based Cells Selection, in *Proc. 2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, Paris, France, 2023, pp. 1245–1253.
6. **B. Soro** and C. Lee, Joint Time–Frequency RSSI Features for Convolutional Neural Network-Based Indoor Fingerprinting Localization, *IEEE Access*, vol. 7, pp. 104892–104899, 2019.
7. **B. Soro** and C. Lee, A Wavelet Scattering Feature Extraction Approach for Deep Neural Network-Based Indoor Fingerprinting Localization, *Sensors*, vol. 19, no. 8, 1790, 2019.
8. **B. Soro** and C. Lee, Performance Comparison of Indoor Fingerprinting Techniques Based on Artificial Neural Networks, in *Proc. TENCON 2018 (IEEE Region 10 Conference)*, Jeju, South Korea, Oct. 28–31, 2018.

Under Review

- **B. Soro et al.**, Generative Initialization of LLMs.

Ongoing Work

- Building biological foundation models by merging heterogeneous foundation-model weights in weight latent space.
- Generative optimization of LLMs.

Talks

Korea AI Expo 2023: Generative Network Hyper-Representations Learning.

Samsung Research Montreal (April 13, 2026): LS-Merge: Merging Language Models in Latent Space.

Teaching

Instructor

Institut Polytechnique Felix Houphouet Boigny

Jan. 2020 – May 2020

Côte d’Ivoire

– Introduction to Machine Learning.

Teaching Assistant

Ajou University

Mar. 2019 – July 2019

Suwon, South Korea

– Signals and Systems.

Industry Experience

Research Intern

Deepauto.AI

Summer 2024

Remote

– Researched weight-space learning and latent-space optimization for pretrained LLMs.

Software Engineering Intern

Aroli Group

2016 (6 months)

Côte d’Ivoire

– Designed a firewall to block gray-route traffic over the GSM SS7 protocol.

Services

NeurIPS 2025: Top Reviewer (registration reward).

Program Committee: AAAI 2026.

Conference Reviewer: ICML, ICLR, NeurIPS, CVPR, AutoConf, WAC, AAAI.

Community Project

Open Low-Resource LLM

2025 – Present

- Goal: democratize LLMs for Senofo by building lightweight models and datasets tailored to the language.
- Outcome: help learners transition from Senofo to French, raise educational attainment, and broaden access to modern technology.

Skills

Programming Languages: MATLAB, C, C++, Python.

Deep Learning Libraries: TensorFlow, PyTorch, Nanotron.

Mathematical Background: Signal Processing, Optimization Theory.

AI: Advanced Topics in Deep Learning.

Scholarships and Grants

Google TPU Grant — \$100,000 in TPU credits + \$50,000 research gift *Apr. 2026 – Present*

INOCORE Postdoctoral Grant, LLMs Innovation Center, KAIST AI *Feb. 2026 – Present*

Korea Government Scholarship (GKS) for international students *Sept. 2020 – Aug. 2023*

Korea Government Scholarship (GKS) for international students *Sept. 2016 – Aug. 2019*

Ivorian Government Scholarship for national students *Sept. 2009 – Aug. 2014*

Fellowships

InnOCORE Postdoctoral Fellowship, KAIST AI *Mars. 2026 – Present*

References

Sung Ju Hwang — Doctoral Advisor and Postdoctoral Supervisor, KAIST AI *Sept. 2022 – Present*

Song Chong — Doctoral Advisor, KAIST AI *Sept. 2020 – Feb. 2026*

Chae Woo Lee — M.Sc. Advisor, Ajou University *Sept. 2017 – Aug. 2019*